Grassroots Design Meets Grassroots Innovation: Rural Design Orientation and Firm Performance

by

Timothy R. Wojan
Oak Ridge Institute for Science and Education Research Ambassadors Program through the National Center for Science and Engineering Statistics

Stephan J. Goetz
Northeast Regional Center for Rural Development, Department of Agricultural Economics, Sociology, and Education, Pennsylvania State University

Zheng Tian
Northeast Regional Center for Rural Development, Department of Agricultural Economics, Sociology, and Education, Pennsylvania State University

Luyi Han
Northeast Regional Center for Rural Development, Department of Agricultural Economics, Sociology, and Education, Pennsylvania State University

CES 24-17 **March 2024**

The research program of the Center for Economic Studies (CES) produces a wide range of economic analyses to improve the statistical programs of the U.S. Census Bureau. Many of these analyses take the form of CES research papers. The papers have not undergone the review accorded Census Bureau publications and no endorsement should be inferred. Any opinions and conclusions expressed herein are those of the author(s) and do not represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. Republication in whole or part must be cleared with the authors.

To obtain information about the series, see www.census.gov/ces or contact Sean Wang, Editor, Discussion Papers, U.S. Census Bureau, Center for Economic Studies, 4600 Silver Hill Road, Washington, DC 20233, CES.Working.Papers@census.gov. To subscribe to the series, please click here.

Abstract

The study of grassroots design—applying structured, creative processes to the usability or aesthetics of a product without input from professional design consultancies—remains under investigated. If design comprises a mediation between people and technology whereby technologies are made more accessible or more likely to delight, then the process by which new grassroots inventions are transformed into innovations valued in markets cannot be fully understood. This paper uses U.S. data on the design orientation of respondents in the 2014 Rural Establishment Innovation Survey linked to longitudinal data on the same firms to examine the association between design, innovation, and employment and payroll growth. Findings from the research will inform questions to be investigated in the recently collected 2022 Annual Business Survey (ABS) that for the first time contains a Design module.

Keyword*:* innovation, research methods, inclusive design, evaluation

JEL Classification: C51, O31, R11, Z11

---

[*] Disclaimer: Any views expressed are those of the authors and not those of the U.S. Census Bureau, the National Center for Science and Engineering Statistics, or the National Science Foundation. The Census Bureau's Disclosure Review Board has reviewed this information product for unauthorized disclosure of confidential information and have approved the disclosure avoidance practices applied to this release. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Numbers 2083 and 2966 (Approval ID: FY23-P2083-R10744). The authors have no competing interests to declare that are relevant to the content of this article.

Acknowledgement: This work was supported in part by the National Science Foundation (NSF), National Center for Science and Engineering Statistics (NCSES), and by the Oak Ridge Institute for Science and Education (ORISE) for the Department of Energy (DOE). ORISE is managed by Oak Ridge Associated Universities (ORAU) under DOE contract number DE-SC0014664. All opinions expressed in this paper are the author's and do not necessarily reflect the policies and views of NSF, DOE, ORAU, or ORISE. This research comes out of a project at the Penn State Federal Statistical Research Data Center that is supported in part by the Department of Agriculture's National Institute of Food and Agriculture under project #2018-67023-27638. This work also was supported in part by the Pennsylvania State University Multistate/Regional Research Appropriations under project #NE2249.

Corresponding author: Timothy R. Wojan, Established Scientist Fellow, Oak Ridge Institute for Science and Education Research Ambassadors Program through the National Center for Science and Engineering Statistics within the National Science Foundation, Oak Ridge, TN, twojan@nsf.gov.

Introduction

Despite increasing evidence that mixed modes of technological and nontechnological innovation characterize the most dynamic businesses (Schmidt and Rammer, 2007; Frenz and Lambert, 2009; Sanchez, 2014; Galindo-Rueda and Millot, 2015; Wojan and Nichols, 2018), little headway has been made in collecting data that would allow a comprehensive analysis of the phenomenon beyond what is available in administrative data on utility patents, design rights or patents, and trademarks (Pinate, et al., 2022). This lack of data is arguably of greater concern in rural areas where both technological and nontechnological innovation are less likely to be documented in administrative data and where the grassroots character of much of this activity can only be assessed through qualitative methods or through survey self-reports. The self-reporting of technological and some forms of nontechnological innovation available in the Community Innovation Survey (CIS) and other Oslo Manual compliant surveys have demonstrated that this grassroots innovation is much more broadly distributed than commonly believed (Wojan, 2022). This paper will investigate the value of extending the study of grassroots innovation to include grassroots design in rural areas—the application of structured, creative processes to the usability or aesthetics of a product that may not be reliant on professional designers. The analysis uses U.S. data on the design orientation of respondents in the 2014 Rural Establishment Innovation Survey (REIS) linked to longitudinal data on the same firms to examine the association between design, innovation, and employment and payroll growth.

Earlier work using these data (Wojan and Nichols, 2018) identified three distinct design orientations of rural firms corresponding to the design ladder construct first proposed by Statistics Denmark (Galindo-Rueda and Millot, 2015). REIS was developed before publication of the design ladder research, but post hoc latent class analysis applied to a group of questions related to design, intellectual property, and intangible investment was used to approximate its rungs. The largest share of firms had no systematic approach to design. Most other firms could be classified as "design last finish:" applying design prior to product launch. The small remaining share of establishments (8% overall, 4.8% in rural firms) were classified as design-integrated, incorporating design through the development process. In addition to the finding that these design-integrated rural firms were also much more likely to engage in more far-ranging and substantive innovation, these firms also tended to concentrate in areas with rich cultural and natural amenities. However, the cross-sectional nature of the study did not allow a direct examination of the association between design orientation and firm performance. All that was possible at the time was a confirmation that counties with design-integrated firms tended to outperform peer counties with respect to higher wages, and that these firms tended to self-report more favourable business indicators over the previous three years relative to design last finish or firms with no systematic approach to design.

The passage of time since the collection of REIS presents the opportunity to examine how the design orientation of firms is associated with objective measures of firm performance such as employment or payroll growth. This longitudinal analysis is made possible by linking REIS to the Longitudinal Business Database maintained by the Census Bureau that uses administrative data, essentially comprising an annual census of employer businesses in the United States. Presentation of the empirical findings is preceded by a review of the compelling preliminary evidence of mixed mode innovation, along with a discussion of how the design orientation of respondents was derived using latent class analysis. The paper concludes with discussion of the limitations of the present analysis and how they may be

addressed with the 2022 Annual Business Survey Design Module data, the value of collecting better data on nontechnological innovation, and the implications for rural innovation policy.

Findings Regarding the Design/Innovation Nexus

The recognition that innovation may happen in the absence of formal science and engineering inputs was the impetus for the creation of the OECD/EU's Oslo Manual in 1992, which provided guidance to member governments on the collection of internationally comparable innovation data (Gault, 2013). However, the inclusion of design activities as part of the innovation process was empirically challenging with test questions on design activity expenditures or other ways to quantify design inputs performing poorly in cognitive testing (Galindo-Rueda and Millot, 2015). The criteria for reporting an innovation were that it be new, or a significant improvement, included the caveat that newness only with respect to aesthetics did not qualify as an innovation.

The EU harmonized the collection of innovation data in the CIS, administered separately by the national statistical offices of member countries. Member countries can include modules beyond the core set of questions, and this allowed Denmark to include questions on the design orientation of firms in the 2010 and 2012 versions of their survey. Instead of attempting to differentiate design activities from other inputs to the innovation process that created problems for respondents in cognitive testing, the objective was to place firms on different rungs of a design ladder that qualitatively defined the orientation and commitment to design as a structured creative process (Galindo-Rueda and Millot, 2015). The first rung is no systematic approach to design that could be characterized as ad hoc design. Design as styling applied to a product before market launch defines the second rung and captures well the assumed contribution of design by economists as a last finish to make a product more aesthetically pleasing. The third rung of the ladder is design as a function or process that is integrated throughout the development of a product but does not treat design as a dominant directing element. The top rung of the ladder is design as strategy, where design is not only integrated throughout product development but is regarded as a critical differentiator for the enterprise. It is these top two rungs that present the possibility that design processes might contribute directly to the utility of an innovation even if through nontechnical inputs.

The analysis of the Statistics Denmark data by the OECD confirms that the design ladder construct produces useful categories for examining the connection between design orientation and innovation (Galindo-Rueda and Millot, 2015). Not surprisingly, firms with no systematic approach to design or who failed to answer the questions made up a large share of the sample (76%). Design as last finish firms characterized 5% of firms with design as process firms making up 12% and design as strategy firms the remaining 7%. However, there was little observed difference in the association between innovation and the last two categories that can be combined and labelled design-integrated firms. Design-integrated firms were more likely to report product, process, marketing, and organizational innovation. In terms of the quality of innovation, design integrated firms also reported a significantly higher share of sales from new or significantly improved products relative to design last finish or no systematic approach to design firms. Design-integrated firms also reported input from design with respect to problem-solving related to new product concepts, work in interdisciplinary teams, and helping to define new business areas (Galindo-Rueda and Millot 2015: pp 27-31).

A post hoc classification using the design ladder categories was put together by applying latent class analysis to design related questions in the 2014 Rural Establishment Innovation Survey (Wojan and

Nichols, 2018). The survey was developed before the design ladder study was published but useful information was available on use of in-house or contracted design services, demonstrated outputs from design work in the form of intellectual property protections, and borrowing funds for intangible investments. Design-integrated establishments were likely to be characterized by all or most of these attributes whereas design last finish firm were only likely to be characterized by in-house or contracted design services. No systematic approach to design establishments possessed none of these attributes. Design-integrated establishments only made up 8% of the sample but 80% of these establishments were substantive or more far-ranging innovators. Design last finish establishment made up 31% of the sample and roughly half were also substantive innovators. Establishments with no systematic approach to design made up 60% of the sample and only 11% were classified as substantive innovators. Most of these design inactive establishments were non-innovators.

These findings of design making substantive contributions to the innovation process reinforce the highly surprising, but little noticed finding out of the National Endowment for the Arts (NEA): that those engaged in design appeared to be direct contributors to invention (Nichols, 2013). A request by the NEA to the U.S. Patent and Trademark Office for a special crosstab of design patent authors and utility patent authors revealed that while only 2% of utility patent authors appeared on design patents, 40% of design patent authors appeared on utility patents. The number of design patents is one-tenth that of utility patents but that would still suggest that authors with a demonstrated design orientation are contributing authors to 4% of inventions. The contribution may be significantly less than that of traditional STEM fields, but it is hard to maintain the assumption that design inputs are ignorable in the processes of invention and innovation. But findings to the contrary demonstrably are.

Other research pointing to the important contribution of nontechnical inputs to innovation have been similarly inert, despite coming from sources more familiar to economists such as the OECD (Frenz and Lambert, 2009) or the Center of Economic Studies at the Census Bureau (Sanchez, 2014). Both analyses apply similar techniques to datasets from selected OECD member countries and the U.S., respectively, to demonstrate the contribution of a factor described as “wider innovation” that includes design and marketing that is important for understanding the innovation process. The synergies of technical and nontechnical innovation are also evident in intellectual property protection where firms rarely rely solely on utility patents but instead usually combine them with design patents or trademark protection (Wojan, 2019).

However, an association between design orientation and the positive measures of self-reported innovation in Oslo Manual compliant datasets is still one step removed from confirming that design is a critical input for understanding economic outcomes. This is because the criterion for reporting innovation is substantive change in what the firm was doing before. There is no requirement that the innovation be successful in the market. The implicit hypothesis that needs to be tested is that design as a structured creative process that replaces design as an *ad hoc* process results in products that are more likely to delight the customer, manifest in better firm performance through time.

Post Hoc Construction of a Design Ladder in REIS

The design-orientation of firms defined by the design ladder construct is the central independent variable of interest in the following analysis so more detail on its derivation and validation is warranted.

Table 1 shows the binary variables used in the latent class analysis to define design-integrated, design last finish, and no systematic approach to design categories, along with the percent of positive responses for each variable by respondents selected into each latent class (Wojan, 2018). The two sets of variables that differentiate the design-integrated from the design last finish class are 1) the two first variables related to any use of design that is either in-house or contracted, and 2) the remaining five variables that suggest a stronger commitment to design in the form of seeking intellectual property protection for nontechnical novelty as well as directing investment to intangibles and being innovation capital constrained. The resulting latent class structure results in a very sharp distinction between those firms with no systematic approach to design that very rarely respond affirmatively to any of these questions and firms that take an explicit approach to design. The differentiation between design-integrated and design last finish is not as sharp with some design last finish firms also seeking nontechnical intellectual property protection or being innovation capital constrained. The latent classes are defined probabilistically so firms identified as design last finish have some positive probability of being classified as either design-integrated or with no systematic approach to design. However, the aggregate response pattern across these three classes is consistent with a decreasing commitment to design going from left to right. The REIS data do not include information on when design inputs are applied during the product development process, but it is reasonable to assume that firm demonstrating the strongest commitment to design are most likely to resemble the design-integrated firms identified explicitly by Statistics Denmark.

	Design Integrated	Design Last Finish	No Systematic Design
Percent of All Establishments	8.11%	31.38%	60.50%
	(0.64%)	(1.01%)	(1.08%)
In-House Design	74.92%*	60.76%*	0.04%
	(4.01%)	(1.88%)	(0.02%)
Contract Design Services	57.64%*	42.69%*	0.79%
	(3.87%)	(1.90%)	(0.39%)
Innovation Capital Constrained	53.78%*	32.63%*	9.00%
	(4.11%)	(1.80%)	(0.89%)
Registered Industrial Design	31.61%*	0.97%*	0%
	(3.51%)	(0.24%)	(0%)
Registered a Trademark	95.28%*	11.11%*	0.84%
	(1.48%)	(1.08%)	(0.23%)
Produced Copyright Eligible Materials	83.48%*	17.15%*	1.26%
	(3.07%)	(1.36%)	(0.27%)
Borrowed Funds for Intangible Investments	19.24%*	8.62%*	0.14%
	(3.24%)	(1.02%)	(0.07%)

Table 1: Responses to Questions Used to Classify Establishments by Design Orientation
*Sources: 2014 Rural Establishment Innovation Survey; Wojan (2018). Notes: * indicates estimate is statistically different from estimate to right at 5% level. Standard error in parentheses.*

Table 2 uses data from the subsample of rural firms in REIS to demonstrate the strong association between grassroots design and grassroots innovation. The innovation classifications are also derived from latent class analysis that differs in an important respect from the Oslo Manual compliant surveys (Wojan, 2018). Affirmatively answering the product innovation question is not enough to be classified as

an innovative firm. Nominal innovators are those firms demonstrating behaviours consistent with a continuous improvement program such as fixing customer complaints consistently, regularly monitoring customer satisfaction, and monitoring employee training. Firms with a rudimentary continuous improvement program can credibly assess incremental innovations. Substantive innovators demonstrate these same behaviours along with behaviours that are consistent with more far-ranging innovation such as producing intellectual property worth protecting, acknowledging past innovation failures, and are likely to be innovation capital constrained. The overwhelming majority of rural design-integrated firms are substantive innovators as are a plurality of design last finish firms. Substantive innovators make up the smallest share of firms with no systematic approach to design. Non-innovators make up a plurality of firms classified as having no systematic approach to design.

	Design Integrated	Design Last Finish	No Systematic Design
Percent of All Establishments	4.76%	28.28%	66.97%
	(0.64%)	(1.01%)	(1.08%)
Substantive Innovator	83.49%	41.31%	9.07%
	(5.92%)	(1.63%)	(0.77%)
Nominal Innovator	11.04%	38.09%	41.00%
	(4.16%)	(1.59%)	(2.49%)
Non-Innovator	5.47%	20.53%	49.93%
	(2.16%)	(1.35%)	(2.38%)

Table 2: Innovation Orientation by Design Orientation for Nonmetropolitan Establishments
Sources: 2014 Rural Establishment Innovation Survey; Wojan (2018).
Note: Standard error in parentheses.

Firms were also asked to report on their performance in the 3 years preceding the survey and Table 3 demonstrates that design-integrated firms generally gave much more positive assessments than either design last finish or firms with no systematic approach to design. The variables most suggestive of superior performance by design-integrated firms are the much higher percent of such firms reporting an increase in market share and entering export markets. Determining if the reported performance advantages are manifest in empirical data is addressed next.

	Design Integrated	Design Last Finish	No Systematic Design
% of 2013 sales from new or improved products	21.72* (1.65)	16.82* (0.787)	9.26 (0.498)
In the past 3 years, the establishment:			
Increased the variety of goods or services offered	90.15%* (1.71%)	78.15%* (1.11)	50.29% (1.02%)
Increased market share or entered new markets	85.46%* (2.47%)	69.86%* (1.19%)	42.31% (1.02%)
Began exporting goods or services	37.52%* (3.09%)	15.27%* (0.88%)	3.76% (0.30%)
Reduced time to respond to customer needs	69.94% (3.01%)	61.24%* (1.29%)	39.76% (0.99%)
Improved flexibility of production or service provision	80.24%* (2.63%)	71.38%* (1.20%)	36.83% (0.96%)
Increased capacity of production or service provision	82.17%* (2.25%)	68.24%* (1.26%)	37.62% (0.96%)
Reduced labour costs per unit output	56.62% (3.34%)	42.13%* (1.29%)	21.49% (0.81%)
Reduced materials and energy required per unit output	49.89%* (3.37%)	34.38%* (1.23%)	13.74% (0.65%)

Table 3: Self-reported performance indicators by design orientation, nonmetropolitan establishments
Sources: 2014 Rural Establishment Innovation Survey; Wojan (2018)
*Notes: * indicates estimate is statistically different from estimate to right at 5% level. Standard error in parentheses.*

Methods

A Census Bureau approved research project proposed linking REIS to the confidential Longitudinal Business Database, providing information on employment, payroll, and continuity of operations through time. These longitudinal data allow a two-stage modelling strategy. In the first stage, we estimate a random coefficient growth model, in which the dependent variable is the level of employment or wage rates of a firm and the regressors are time trend and its quadratic term. The random coefficient growth model can be written as

$$\begin{aligned} y_{it} &= \beta_{0i} + \beta_{1i}t + \beta_{2i}t^2 + \varepsilon_{it} \\ \beta_{0i} &= \pi_{00} + b_{0i} \\ \beta_{1i} &= \pi_{10} + b_{1i} \end{aligned} \tag{1}$$

$$\beta_{2i} = \pi_{20} + b_{2i}$$

in which each firm i would have its own growth parameters $\beta_{0i}, \beta_{1i}, \text{and } \beta_{2i}$. b_{0i}, b_{1i} and b_{2i} are the random effects assumed to be jointly distributed with a multivariate normal distribution, i.e., $(b_{0i}, b_{1i}, b_{2i})' \sim MVN(\mathbf{0}, \boldsymbol{\Sigma})$. ε_{it} is the within-group error that is assumed to follow a first-order autoregressive process, i.e., $\varepsilon_{it} = \rho\varepsilon_{it-1} + \xi_{it}$, where ξ_{it} is a white noise. t = 0, …, 5 represents the years from 2014 to 2019. With the estimated coefficient for each firm, we can compute the growth rate ($\mathrm{d}y_{it}/\mathrm{d}t$) for each time t = 1, …, 5 as $v_{it} = \hat{\pi}_{10} + \hat{b}_{1i} + 2(\hat{\pi}_{20} + \hat{b}_{2i})t$. v_{it} is then fed into the second-stage quantile regression model as the dependent variable. That is,

$$Q_\tau(v_{it}|X_{it}) = X_{it}'\theta(\tau), \text{ for } \tau \in (0,1), t = 1, \ldots, 5 \quad (2)$$

where τ represents the quantiles of v_{it}, Xit include the dummy variable for design-integrated orientation and other firm specific variables, and the coefficients $\theta(\tau)$ vary along the quantile. The use of quantile regression is indicated in those cases where right-tail phenomena are thought to dominate which is the case with both innovation and the relatively rare design-integrated orientation. It is likely that the estimate of mean or median relationships will be uninformative but that tests at the 75th or 90th percentiles will be required to detect the hypothesized mixed mode kick. These percentiles were defined prior to estimation based on earlier analysis of firm performance related to substantive innovation (Tian, et al., 2023).

Results

Table 4 presents results from the ordinary least squares (OLS) and quantile regressions for the entire sample including both firms in metropolitan and nonmetropolitan areas. As expected, no design orientation effect on employment is detected at the mean (OLS) or median (Tau = 0.5). At the 75th percentile, employment growth effects are detected for both design last finish and design integrated orientation. The estimate of the design last finish effect is more precise at the 75th percentile, most likely due to the greater share of firms characterized with this design orientation relative to the design integrated orientation. However, at the 90th percentile, the coefficients are estimated with similar precision. The magnitude of the points estimates suggests that the effect of a design-integrated orientation is roughly 3 times that of a design last finish orientation in the 90th percentile of employment growth, or a growth rate of 1.65% versus 0.51%. While we cannot ascribe a causal effect to design orientation—i.e., we do not know if firms currently pursuing *ad hoc* design would be likely to experience similar employment growth effects by adopting a systematic approach to design—the association does strongly suggest that nontechnological innovation embodied in design is important to firm performance.

Variable	OLS	Tau = 0.5	Tau = 0.75	Tau = 0.9
intercept	-0.0246 *	-0.0291 **	7e-04	0.0082
design last finish	9e-04	0.0011	0.0026 ***	0.0051 ***
design integrated	0.0026	0.0019	0.0045 *	0.0165 ***
log(Employees in 2014)	0.0159 ***	0.0203 ***	0.0181 ***	0.0157 ***
firm age in 2014	-2e-04 **	-2e-04 ***	-4e-04 ***	-6e-04 ***
Num. obs.	5,100	5,100	5,100	5,100

Table 4. Estimated OLS and quantile regression coefficients for employment growth

Sources: 2014 Rural Establishment Innovation Survey; 2014-2019 Longitudinal Business Database. FSRDC Project Number 2086, Census Bureau Disclosure Approval Number FY23-P2083-R10744.
*Note: *, **, and *** indicate estimates are statistically significant at the 0.10, 0.05, and 0.01 levels, respectively.*

In contrast, the effects of design orientation on wage growth shown in Table 5 are not detected in the top quantile, and no effects of design last finish orientation are detected. Instead, an effect of the design integrated orientation is detected in the middle of the wage growth distribution around the 50th and 75th percentiles. Without a strong conceptual argument for why such effects might be expected in the middle of the wage growth distribution, it is prudent to conclude that no verdict on the effects of design orientation on wage growth is available. Revisiting this question when the much larger 2022 ABS Design Module data are available could examine hypotheses related to the characteristics of industries that design-integrated firms tend to concentrate in. But such a sectoral analysis is not possible given the relatively small size of REIS, creating implicit samples that would not pass disclosure avoidance thresholds.

Variable	OLS	Tau = 0.5	Tau = 0.75	Tau = 0.9
intercept	0.0035	0.0039	0.0171	0.0205
design last finish	0.0012	2e-04	2e-04	-0e+00
design integrated	-0.0032	0.0047 **	0.0045 **	0.0017
log(PAY/WORKER(2014))	0.0146 ***	0.0204 ***	0.016 ***	0.0103 ***
firm age in 2014	1e-04	-0e+00	-1e-04 ***	-2e-04 ***
Num. obs.	5,100	5,100	5,100	5,100

Table 5. Estimated OLS and quantile regressions coefficients for wage growth

Sources: 2014 Rural Establishment Innovation Survey; 2014-2019 Longitudinal Business Database. FSRDC Project Number 2086, Census Bureau Disclosure Approval Number FY23-P2083-R10744.
*Note: *, **, and *** indicate estimates are statistically significant at the 0.10, 0.05, and 0.01 levels, respectively.*

Table 6 presents results regarding employment growth limited to the nonmetro sample. Nonmetro design last finish firms are qualitatively indistinguishable from the full sample. A statistically significant effect on employment growth is apparent at both the 75th and 90th percentiles. In contrast, only a marginally significant result at the 90th percentile characterizes nonmetro design integrated firms. However, the loss of precision is not surprising given that fewer than five percent of nonmetro firms were classified as design intensive and that the sample is 25% smaller. The magnitude of the point estimate is potentially more informative given these limitations, and nonmetro design-integrated firms demonstrated growth rates roughly twice that of design last finish firms. A strict frequentist may deem the design-integrated finding inconclusive by failing to meet the 0.05 significance threshold.

Practitioners using these data to inform decisions may be more comfortable with the concluding that the more substantive design orientation of rural firms is associated with faster employment growth.

Variable	OLS	Tau = 0.5	Tau = 0.75	Tau = 0.9
intercept	-0.0308 **	-0.0313 **	-6e-04	0.0098
design last finish	2e-04	0.0015	0.0031 ***	0.0058 ***
design integrated	0.0012	0.0025	9e-04	0.0111 *
log(Employees in 2014)	0.0163 ***	0.0204 ***	0.0181 ***	0.0157 ***
firm age in 2014	-1e-04 **	-1e-04 ***	-3e-04 ***	-5e-04 ***
Num. obs.	3,900	3,900	3,900	3,900

Table 6. Estimated OLS and quantile regression coefficients for nonmetro employment growth
Sources: 2014 Rural Establishment Innovation Survey; 2014-2019 Longitudinal Business Database. FSRDC Project Number 2086, Census Bureau Disclosure Approval Number FY23-P2083-R10744.
*Note: *, **, and *** indicate estimates are statistically significant at the 0.10, 0.05, and 0.01 levels, respectively.*

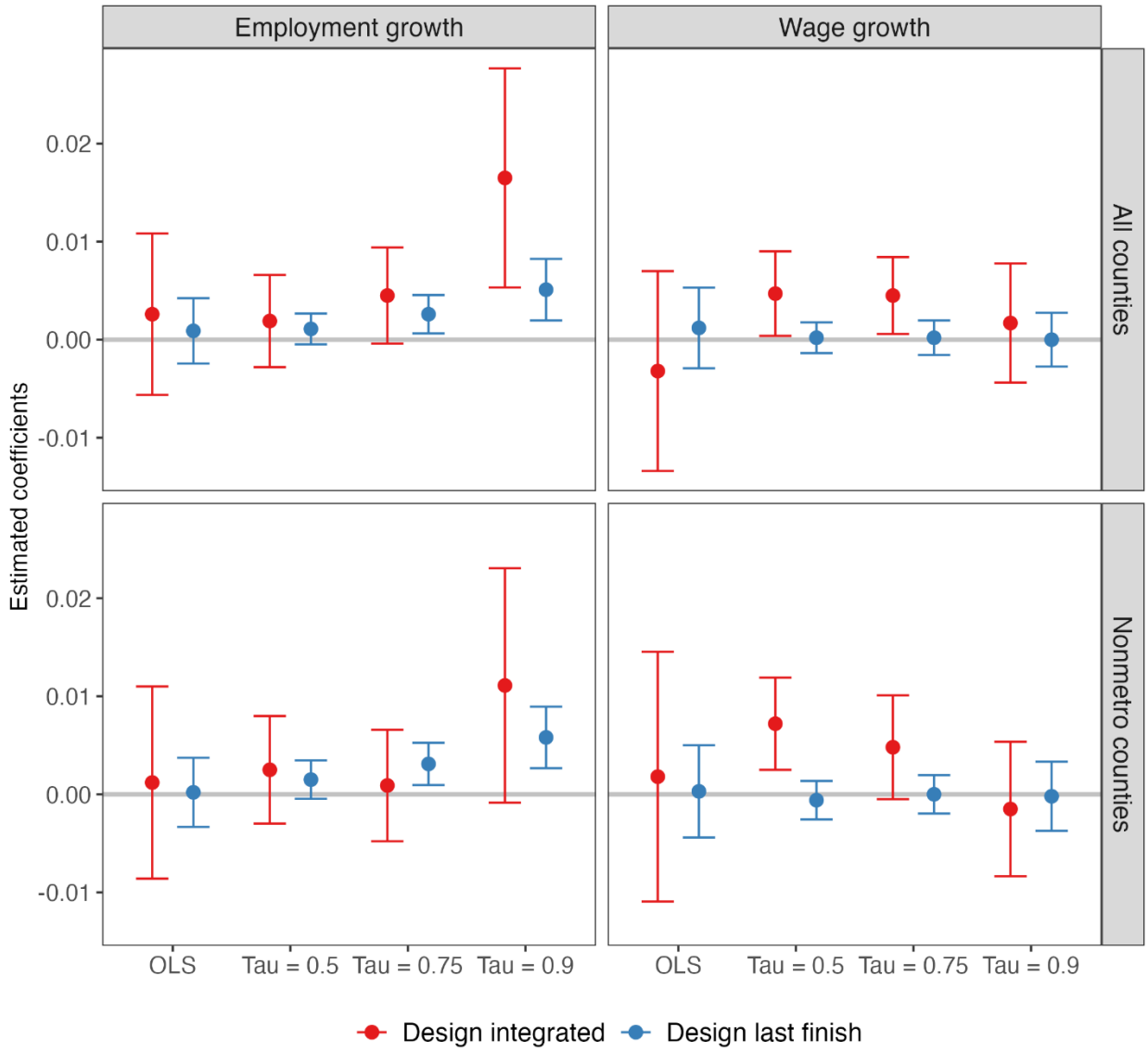
Table 7 is provided for completeness and demonstrates a similar median quantile wage growth effect for nonmetro design integrated firms observed in the full sample. The nonmetro sample also demonstrates the same wage divergence phenomenon observed in the full sample where higher wage firms demonstrate faster wage growth throughout the wage growth distribution. The phenomenon is also observed in the employment growth Table 4 and Table 6 where larger firms also appear to have faster growth rates throughout the employment growth distribution. Of the two phenomena, the employment growth effect is less surprising given that some firms do not pursue growth and this type of firm will be more concentrated in the smallest firm size class by definition. The wage phenomenon may be more indicative of skill-biased technical change, greater bargaining power of organized labour that tend to have higher wages, or lower-skill firms subject to more intensive import competition.

Variable	OLS	Tau = 0.5	Tau = 0.75	Tau = 0.9
intercept	0.0056	0.008	0.0238 **	0.0567
design last finish	3e-04	-6e-04	0e+00	-2e-04
design integrated	0.0018	0.0072 ***	0.0048 *	-0.0015
log(PAY/WORKER(2014))	0.0167 ***	0.0222 ***	0.0165 ***	0.0114 ***
firm age in 2014	1e-04	-0e+00	-1e-04 *	-2e-04 ***
Num. obs.	3900	3900	3900	3900

Table 7. Estimated OLS and quantile regression coefficients for nonmetro wage growth *Sources: 2014 Rural Establishment Innovation Survey; 2014-2019 Longitudinal Business Database. FSRDC Project Number 2086, Census Bureau Disclosure Approval Number FY23-P2083-R10744. Note: *, **, and *** indicate estimates are statistically significant at the 0.10, 0.05, and 0.01 levels, respectively.*

Finally, we visualize the comparison of estimated coefficients on design orientation variables in all cases in Figure 1. It is easy to see that the employment growth rates at the 90th quantile are significantly higher for firms with integrated design processes than all other cases.

Figure 1. Comparison of estimated difference in employment and wage growth among firms with different forms of design-oriented innovations.



Sources: 2014 Rural Establishment Innovation Survey; 2014-2019 Longitudinal Business Database. FSRDC Project Number 2086, Census Bureau Disclosure Approval Number FY23-P2083-R10744.

Discussion

Directly measuring the association between the design orientation of rural firms and firm performance has important implications for rural development. Non-technological inputs provided by design to innovation are rarely discussed in the economics of innovation literature, and even rarer in the context of rural innovation. The conventional perception of design—done by specialized design consultancies or corporate design units—is an urban phenomenon, arguably most prevalent in global cities (Reimer et al. 2008). Building on the earlier findings that design as a structured, creative process is in fact present in rural areas (Wojan and Nichols, 2018), examination of the employment and payroll trajectories of design-integrated firms relative to firm where design is ad hoc confirms a strong association with firm performance. While the present analysis fails to definitively establish a causal link between an explicit focus on design and firm performance, additional observational studies are unlikely to reduce this uncertainty given the complexity of the processes of design, innovation, and value creation. Instead, pilot projects directed to increasing the design capability of rural firms that are predisposed to exploit such opportunities would provide much more valuable information on the efficacy of broadening grassroots design. The fact that resources used in design—imagination, empathy, artisanal tinkering, divergent thinking, learning by making/drawing—are more footloose than those for technological innovation suggest that pilot rural design initiatives may be highly feasible.

The findings are also relevant to the analysis of the 2022 ABS that includes a Design module designed to elicit more concrete information than that available in REIS. Questions are asked about the how design is integrated into firm strategy, along with questions on the formal commitment to design activities in the form of a design budget, the occupational groups involved in design, and how design work is organized. These data will address the major limitation of the current analysis that infers placement on the design ladder with characteristics that may conflate innovation and design. In contrast, data from the Design module will be able to construct a more robust design ladder using explicit information on how design is used in the firm. Questions on trends regarding the importance of design and anticipated allocation of design resources will provide insight on where design is emerging and where it may be waning. Preliminary analysis will suffer from the same cross-sectional limitations of the initial REIS analysis but will benefit from a companion module on globalization activities of firms, and a sample size 30 times that of REIS. Findings from REIS and discussion of the 2022 ABS Design module may also inform the value of including a design module in the EU CIS, which to date has only been fielded in Denmark.

References

Anderson, M.L. and Magruder, J. 2017. *Split-sample strategies for avoiding false discoveries*. NBER Working Paper No. 23544. https://www.nber.org/papers/w23544

Brown, T. 2019. *Change by Design, Revised and Updated: How Design Thinking Transforms Organizations and Inspires Innovation.* New York: HarperCollins.

Galindo-Rueda, F. & Millot, V. 2015.Measuring design and its role in innovation. OECD Science, Technology and Industry Working Paper 2015/01. Paris: OECD Publications.

Gault, F. 2013. The Oslo Manual. In F. Gault (Ed.), *Handbook of Innovation Indicators and Measurement* (pp. 41-59). London: Edward Elgar,

Frenz, M. & Lambert, R. 2009. Exploring non-technological and mixed modes of innovation across countries. In Innovation in Firms: A Microeconomic Perspective (pp. 69-110). Paris: OECD Publishing, https://doi.org/10.1787/9789264056213-4-en.

Nichols, B. 2013. Valuing the art of industrial design: a profile of the sector and its importance to manufacturing, technology, and innovation. National Endowment for the Arts Research Report #56. Washington, DC: National Endowment for the Arts.

Pinate A.C., Faggian A., Di Berardino C., and Castaldi C. (2022). The heterogenous relationship between migration and innovation: Evidence from Italy. *Industry and Innovation* 30(3):336-60. DOI: 10.1080/13662716.2022.2138279

Reimer, S., Pinch, S. and Sunley, P., 2008. Design spaces: agglomeration and creativity in British design agencies. *Geografiska Annaler: Series B, Human Geography*, 90(2), pp.151-172.

Sanchez, J. 2014. Non-technological and mixed modes of innovation in the United States: Evidence from the Business Research and Development Innovation Survey, 2008-2011. US Census Bureau Center for Economic Studies Paper No. CES-WP 14-35.

Schmidt, T. & Rammer, C. 2007. Non-technological and technological innovation: Strange bedfellows? ZEW-Centre for European Economic Research Discussion Paper No. 07-052. Available at SSRN: https://ssrn.com/abstract=1010301 or http://dx.doi.org/10.2139/ssrn.1010301

Tian, Z., Wojan, T.R., & Goetz, S.J. 2023. Growth trajectories of innovative rural firms. Western Regional Science Council meeting in Big Island, Hawaii, February 15-18.

Wojan, T.R.; National Center for Science and Engineering Statistics (NCSES). 2022. The geography of self-reported innovation: Results from the 2017 Annual Business Survey. Working Paper NCSES 22-206. Alexandria, VA: National Science Foundation. Available at https://ncses.nsf.gov/pubs/ncses22206.

Wojan, T.R. 2018. Supporting Information S1: Latent class analysis using Rural Establishment Innovation Survey data, validation tests, and regression results with industry controls, for Wojan and Nichols 2018. *PLoS ONE* 13(2): https://doi.org/10.1371/journal.pone.0192962.s001

Wojan, T.R. & Nichols, B. 2018. Design, innovation, and rural creative places: Are the arts the cherry on top, or the secret sauce? *PLoS ONE* 13(2): e0192962. https://doi.org/10.1371/journal.pone.0192962.